

ECS-323 – Control Systems

Vision-Guided Autonomous Mobile Dustbin

Mid-Term Project Report

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Abstract

This project proposes the development of a vision-guided autonomous mobile dustbin for detecting the trajectory of thrown waste, predicting its landing location, and subsequently moving to the required location for waste A collection, the entire system is to combine computer vision, The prediction of trajectory, the study of mobile robot kinematics, and closed-loop control math.

A MATLAB program is used for the analysis. The trajectory involves the addition of measurement noise to account for imperfect sensing, A polynomial model is then used to fit the measured trajectory. The future ground-intersection point is then estimated as the predicted landing location.

The present implementation sets up the sensing and prediction subsystem. The mobile robot control, three-wheel kinematic model, feedback controller, Automatic lid mechanisms are also intended for the later stages.

1 Introduction

Conventional dustbins require the user to manually approach the bin and Get rid of waste. The project in question is aimed at developing an autonomous mobile dustbin that can respond to thrown waste by estimating where the waste will First it moved to the land and then placed itself at the required position.

The complete proposed system consists of the following major stages:

The use of a vision-based sensing system for detecting waste that has been thrown. The motion and trajectory of the waste can be estimated. The prediction of where the landing will take place. The calculation of the movement needed by the mobile dustbin. The three-wheel mobile platform is controlled using a closed-loop system. The dustbin lid opens and closes automatically.

For the present mid-term evaluation, only the first three stages have been implemented in MATLAB.

2 Objectives

The objectives of the current project milestone are:

Create a mathematical model of the trajectory of waste that has been thrown. Simulate the waste movement using MATLAB.

1. Add measurement noise to show imperfect sensing. Estimate the trajectory based on the noisy measurements. Estimate when the waste will land in the future.
2. Find the matching horizontal landing spot.
3. Establish a reliable reference point for the future mobile-dustbin control system.

3 Proposed System Architecture

The final proposed system is a closed-loop autonomous system. The conceptual architecture is shown below.

Block Diagram For the project

4 Mathematical Model of Waste Trajectory

The thrown waste is initially modeled as a projectile moving in a two dimensional plane.

Let the initial horizontal position be x_0 , initial height be z_0 , horizontal velocity be v_{x0} , and vertical velocity be v_{z0} .

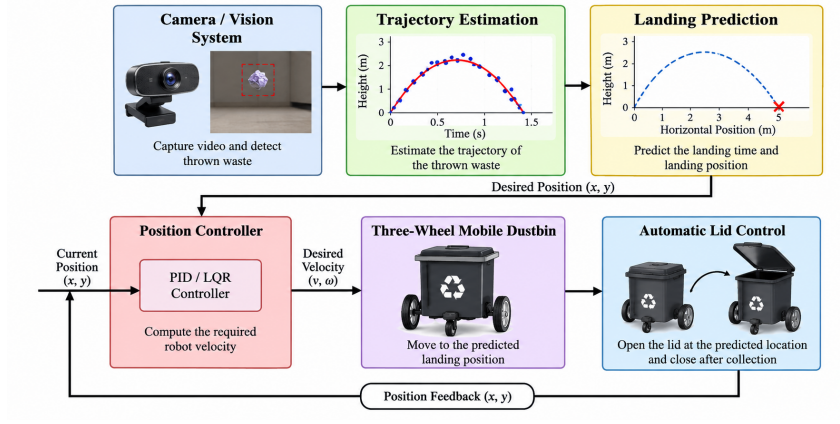


Figure 1: Block Diagram

The horizontal position is given by

$$x(t) = x_0 + v_{x0}t. \quad (1)$$

The vertical position is given by

$$z(t) = z_0 + v_{z0}t - \frac{1}{2}gt^2, \quad (2)$$

where

$$g = 9.81 \text{ m/s}^2 \quad (3)$$

is the gravitational acceleration.

The landing time is obtained when the waste reaches the ground:

$$z(t_{\text{land}}) = 0. \quad (4)$$

Therefore,

$$z_0 + v_{z0}t_{\text{land}} - \frac{1}{2}gt_{\text{land}}^2 = 0. \quad (5)$$

Once the landing time is obtained, the corresponding horizontal landing position can be calculated as

$$x_{\text{land}} = x_0 + v_{x0}t_{\text{land}}. \quad (6)$$

This landing position will later serve as the desired position for the mobile dustbin.

5 Simulated Measurement Model

In a practical vision-based system, measurements are affected by camera resolution, image processing errors, lighting conditions, and other sources of uncertainty.

To represent this behavior in the simulation, Gaussian measurement noise is included in the actual trajectory.

The measured horizontal position is modeled as

$$x_m(t) = x(t) + \eta_x(t), \quad (7)$$

and the measured vertical position is modeled as

$$z_m(t) = z(t) + \eta_z(t), \quad (8)$$

where η_x and η_z stand for measurement noise.

This allows the simulation to reproduce a more realistic sensing condition Rather than supposing that the measurements are perfect.

6 Trajectory Estimation

The underlying trajectory is estimated using the noisy measurements.

A second-order polynomial is fitted to the measured vertical trajectory:

$$z(t) = at^2 + bt + c. \quad (9)$$

The estimated landing time is obtained by solving

$$at^2 + bt + c = 0. \quad (10)$$

Among the resulting roots, the positive future intersection with the ground is chosen as the estimated time of landing.

The horizontal trajectory is estimated using a first-order polynomial:

$$x(t) = mt + c_x. \quad (11)$$

The predicted horizontal landing position is then obtained by evaluating this estimated model at the predicted landing time:

$$x_{\text{pred}} = x(t_{\text{land,pred}}). \quad (12)$$

7 MATLAB Implementation

The current implementation is divided into three MATLAB files.

Table 1: MATLAB source files and their functions

File	Function
<code>main.m</code>	Main execution and visualization
<code>simulate_trajectory.m</code>	Trajectory and noisy measurement generation
<code>predict_landing_point.m</code>	Landing-time and landing-position prediction

The complete simulation can be executed using the single command:

```
main
```

This makes the current milestone reproducible without requiring manual modification of the source code.

8 Simulation Parameters

The current MATLAB simulation uses the following parameters.

Table 2: Simulation parameters

Parameter	Value
Gravitational acceleration, g	9.81 m/s ²
Initial horizontal position, x_0	0 m
Initial height, z_0	2 m
Initial horizontal velocity, v_{x0}	3.5 m/s
Initial vertical velocity, v_{z0}	5.0 m/s
Measurement noise standard deviation	0.06 m
Number of simulated measurements	80

9 Simulation Results

The MATLAB simulation produces the true trajectory, noisy measurements, estimated trajectory, and predicted landing point.

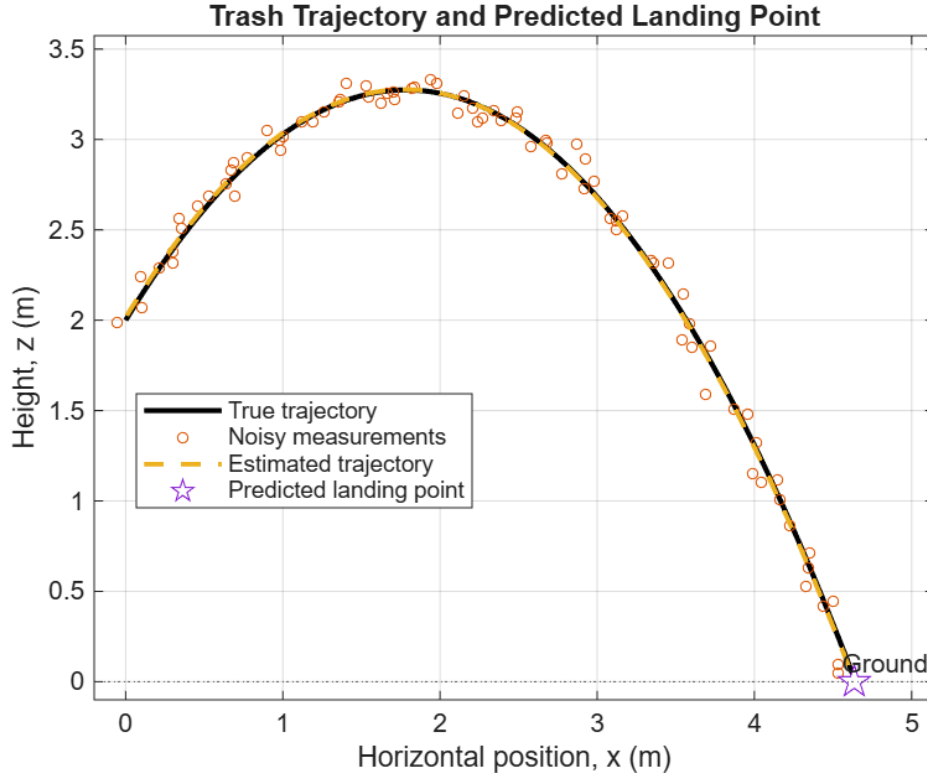


Figure 2: Simulated trash trajectory, noisy measurements, estimated trajectory, and predicted landing point.

The simulation demonstrates that the landing position can be estimated from noisy trajectory measurements. The predicted landing location obtained from the MATLAB implementation provides the reference location that will subsequently be supplied to the mobile-dustbin control system.

The current implementation prints the predicted landing time and horizontal landing position directly in the MATLAB command window.

10 Current Project Status

The following components have been completed for the current milestone:

A model of the waste trajectory based on projectiles.

- MATLAB trajectory simulation.
- Simulated noisy measurements.
- Polynomial trajectory estimation.
- Landing-time prediction.

- Landing-position prediction. The visualization of the predicted landing point in MATLAB.

The following components are intentionally reserved for future milestones:

A three-wheel mobile robot model.

- Three-wheel inverse kinematics.
- Position feedback control.
- PID/LQR controller.
- Closed-loop dustbin motion. Item: rejection of disturbances and comparison of the controllers.
- Automatic lid control.
- Complete autonomous simulation.

11 Future Work

The next stage of the project will use the predicted landing location as The input used as a reference for the autonomous mobile dustbin.

The planned control architecture is

$$\text{Predicted Landing Point} \rightarrow \text{Position Error} \rightarrow \text{Controller} \rightarrow \text{Three-Wheel Robot} \rightarrow \text{Position Feedback.} \quad (13)$$

A suitable controller such as PID or LQR will be investigated for reducing the position error.

The three-wheel geometry like rickshaw will then be incorporated.

Finally, an automatic lid-control mechanism will be added so that the lid opens when the dustbin reaches the required trash pickup position and closes When the waste has been received.

12 Conclusion

This milestone establishes the sensing and prediction component of the A mobile dustbin guided by vision.

A model based on the principles of projectile motion was created using MATLAB to simulate waste that is thrown. Measurement noise was introduced to simulate realistic

sensing conditions, and polynomial fitting was used to estimate the trajectory and predict the landing point.

The predicted landing position provides the required reference for the This study therefore constitutes a future autonomous motion-control subsystem. the foundation for the subsequent development of the three-wheel mobile a dustbin together with its closed-loop control system.